Process book

Team

Team Name: TheVizG.O.A.T
Team Leader: Kunhao Ni
Team members:

Qixiang(David) Fan - qfan49@gatech.ed
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Team agreement

- All code should be clearly documented. Team members are expected to include sufficient comments and documentation to explain the purpose, implementation steps, and important design decisions so that other members can understand and maintain the code.
- Workload and working hours should be distributed as evenly as possible among all team members. Individual tasks may differ depending on members' experience and strengths, but we will aim to ensure that everyone contributes fairly to the project.
- We will use Git for version control and collaborative development. Team members should regularly commit their work and use appropriate branches and pull requests when integrating major changes.
- We will adopt a hybrid collaboration model. For day-to-day communication, we will primarily use WeChat and email to share updates, ask questions, and coordinate tasks.
- We will hold a Zoom meeting every Friday to review the progress made during the week, discuss any issues or design decisions, and assign or adjust tasks for the following week.
- Major project decisions, including visualization design, data processing approaches, and implementation choices, will be discussed among team members before being finalized.
- If a team member consistently fails to complete assigned tasks and refuses to communicate about delays, the issue will first be raised directly with them. If the issue continues, it will be discussed with the whole team and, if still unresolved, escalated to the instructor/TA.

Signatures: Kunhao Ni, Yurun Zhu, Qixiang Fan, Zhaorui Ding

Date: Sep 4, 2026

Project Proposal

Our team is aiming to analyze data for basketball. Potential visualization might include a multi-dimensional ability graph for players.

Possible datasource:

- https://www.kaggle.com/datasets/wyattowalsh/basketball
- nba.com

NBA Historical Data Visualization

We aim to build an interactive visualization page that shows how the NBA tactics and playing styles have changed over time. We plan to visualize, explore and analyze the data from three main perspectives: shot locations distribution, league-wide playing style, and player or team performance (including comparisons).

First, we want to analyze how shot selection has changed over time during different NBA eras. For example, we could create visualizations such as heatmaps on a basketball court graph to show where the shots were taken, and what the rate of success differed from each other. We could also add a season slider to make the heatmap displaying how the league has gradually changed from mid-range shots Era toward three-pointers/near rim shots Era. Since there are tons of NBA stats datasets available on Kaggle, it makes it easier for us to focus on the implementation.

We also plan to visualize how the playing style of the NBA has changed over time. For example, overall points per game, three-pointers proportion, assists over points, rebounds per game, free throws proportion, average time per offensive possession, etc.. This would help us compare and analyze how NBA strategies have evolved, over decades.

Finally, we could allow users to compare different players or teams across seasons. Users could select several players or teams and compare statistics such as points, true shooting percentage, or other performance metrics over time.

For the data, we plan to mainly use Kaggle NBA datasets and NBA.com Stats with the API it provides, since both provide a large amount of historical player, team, and shooting data.

Week 3 - MAP

Data Collection and Project Plan

Part I - Audience Analysis

Generally, we have three main groups of audiences: first NBA fans and enthusiasts. Second sports data analyst who might use the data to help them manage the team or guys the training of the team. Third, PE teachers who use the data as every life example to show the trends and techniques in modern basketball.
We choose the first group: the fans and enthusiasts as our audience. According to observation on social media, they have more interest in different patterns of data other than usual analytics metrics, and that might enable us to dig and explore more features from the data.

Part II - Target Audience

NBA fans usually know most of the basketball rules. They have comprehensive concepts and understanding of points, fouls, attack and defense, etc. They know the match schedules and the teams of the NBA. They usually know most of the star players and have one or several of their favorites. A lot of NBA fans also have understanding and knowledge of other players, including those currently playing and those already retired. And they usually know the history of matches. They are usually interested in who is the best player (either currently or in history), which team is competitive or has the potential of winning the next season, detailed comparisons between two players, for example, Shaquille O'Neal versus Hakeem Olajuwon, in different metrics or dimensions. They are also interested in how the playing style has changed over time.

Their visualization literacy is that usually they can understand charts like line charts, scatter plots, pie charts, bar charts, and radar charts, and they can use these visualizations to enhance their understanding of a team, a player, or a trend. Other than that, since they are usually familiar with the court, they can also understand visualizations that are based on the space of the court, for example, points distribution, etc.

Part III - Potential Questions

Given these facts, we would like to present and visualize the data and information from a higher level to a detailed or lower level. For example, first show the playing style or points distribution of all teams over the season, then show a single team's evolution, or some comparisons between the styles of different teams, and take the star player or leading player of their teams as an example, show their playing data, and make some analysis and visualization. Some questions that our audience may have for our data include:

Questions:

1. Where on an NBA court are fights most likely to break out?
2. How long are the 10 longest three-pointers ever made?

3. Is Kardashian Curse real? Can we see it from the player stats?
4. Is there a connection between a team's geographic location and its achievements?
5. Does being taller lead to better performance in the NBA?
6. How has the average points per game changed over all the seasons?
7. How have two-pointers and three-pointers changed over the seasons? And what does that indicate about the playing styles of the NBA?
8. Which teams rely more on three-pointers in recent seasons, or over this decade? Does more 3-pointers mean more win rate?
9. What is the average distribution of points over the four quarters of each game in recent seasons? What about in history? How has such proportion changed?
10. What is the median season ranking of championship teams? Roughly how high is the "dark horse" rate (i.e., the frequency of unexpected championship teams)?
11. Is there a noticeable skew in the role distribution between a team's first and second leading scorers? How many championship teams clearly show a strong "single-core carry" tendency, versus superteams with multiple stars or stacked rosters?
12. How are scoring, defense, and assists distributed among the five positions on the court, and how has this trend changed throughout history? For each of the five positions, which dimension stands out on average?
13. Who is the G.O.A.T. on each dimension/ position?
14. For each of the five positions, which dimension stands out on average?
15. How is the scoring efficiency of different NBA stars, especially FMVP or MVP winners? Are they genuinely impactful, or just stat-padders?

Part IV - Data Summary

We planned to use two sources as main data: the NBA Basketball Database on Kaggle and NBA Stats data accessed through the Python package nba_api.

The Kaggle database provides related tables containing historical player, team, game, and shooting data. The nba_api allowed us to collect supplementary records when needed.

We will use nba_api to collect supplementary statistics when needed. These include player performance, individual shot details, and advanced measures such as pace, offensive rating, and true shooting percentage. The selected attributes are listed in the NBA API Attributes table:

Attribute / field	Description	Data type	Table / endpoint
player_id	Unique player identifier.	Categorical	dim_player

full_name	Player name.	Categorical	dim_player
position	Listed basketball position.	Categorical	dim_player
height	Player height; convert text to inches or centimeters.	Quantitative after conversion	dim_player
weight	Player weight in pounds.	Quantitative	dim_player
team_id	Unique team identifier.	Categorical	dim_team
full_name	Full team name.	Categorical	dim_team
abbreviation	Short team label, such as LAL.	Categorical	dim_team
city	Team city.	Geographic / categorical	dim_team
state	Team state.	Geographic / categorical	dim_team
conference	Eastern or Western Conference.	Categorical	dim_team
division	Team division.	Categorical	dim_team
game_id	Unique game identifier.	Categorical	dim_game
game_date	Date the game was played.	Temporal	dim_game
season_year	Season label, ordered chronologically.	Temporal / ordinal	dim_game
season_type	Regular season, playoffs, or other competition type.	Categorical	dim_game
home_team_id	Identifier of the home team.	Categorical	dim_game

visitor_team_id	Identifier of the visiting team.	Categorical	dim_game
pts	Points scored.	Quantitative	fact_team_game
fgm	Field goals made, including two- and three-pointers.	Quantitative	fact_team_game
fga	Field goal attempts, including two- and three-pointers.	Quantitative	fact_team_game
fg_pct	Proportion of field goal attempts made.	Quantitative	fact_team_game
fg3m	Three-pointers made.	Quantitative	fact_team_game
fg3a	Three-point attempts.	Quantitative	fact_team_game
fg3_pct	Proportion of three-point attempts made.	Quantitative	fact_team_game
ftm	Free throws made.	Quantitative	fact_team_game
fta	Free throw attempts.	Quantitative	fact_team_game
ft_pct	Proportion of free throw attempts made.	Quantitative	fact_team_game
oreb	Offensive rebounds.	Quantitative	fact_team_game
dreb	Defensive rebounds.	Quantitative	fact_team_game
reb	Total rebounds.	Quantitative	fact_team_game
ast	Assists.	Quantitative	fact_team_game
stl	Steals.	Quantitative	fact_team_game

blk	Blocked shots.	Quantitative	fact_team_game
tov	Turnovers.	Quantitative	fact_team_game
pf	Personal fouls.	Quantitative	fact_team_game
pts_qtr1	Team points scored in quarter 1.	Quantitative	fact_team_game
pts_qtr2	Team points scored in quarter 2.	Quantitative	fact_team_game
pts_qtr3	Team points scored in quarter 3.	Quantitative	fact_team_game
pts_qtr4	Team points scored in quarter 4.	Quantitative	fact_team_game
period	Quarter number, followed by overtime periods.	Ordinal	fact_shot_chart
minutes_remaining	Minutes remaining in the period.	Quantitative (duration)	fact_shot_chart
seconds_remaining	Seconds component of the remaining period time.	Quantitative (duration)	fact_shot_chart
shot_type	Two-point or three-point attempt.	Categorical	fact_shot_chart
action_type	Shooting action, such as a jump shot or layup.	Categorical	fact_shot_chart
shot_zone_basic	Broad court region of the shot.	Categorical / spatial	fact_shot_chart
shot_zone_area	Directional court area of the shot.	Categorical / spatial	fact_shot_chart
shot_zone_range	Shot distance category; order distance bands where applicable.	Ordinal / categorical	fact_shot_chart
shot_distance	Shot distance in feet.	Quantitative	fact_shot_chart

loc_x	Horizontal court coordinate; verify coordinate scale.	Quantitative / spatial	fact_shot_chart
loc_y	Vertical court coordinate; verify coordinate scale.	Quantitative / spatial	fact_shot_chart
shot_made_flag	Made (1) or missed (0).	Categorical (binary)	fact_shot_chart
wins	Season wins.	Quantitative	fact_standings
losses	Season losses.	Quantitative	fact_standings
win_pct	Proportion of games won.	Quantitative	fact_standings
conf_rank	Rank within the conference.	Ordinal	fact_standings
div_rank	Rank within the division.	Ordinal	fact_standings

Beside that, we are also considering use following sources to supplement our main datasets once we need:

Additional Potential Data Sources

1. **Basketball Reference**
2. **NBA Play-by-Play & Shot Data**
3. **PBP Stats**
4. **NBA History**

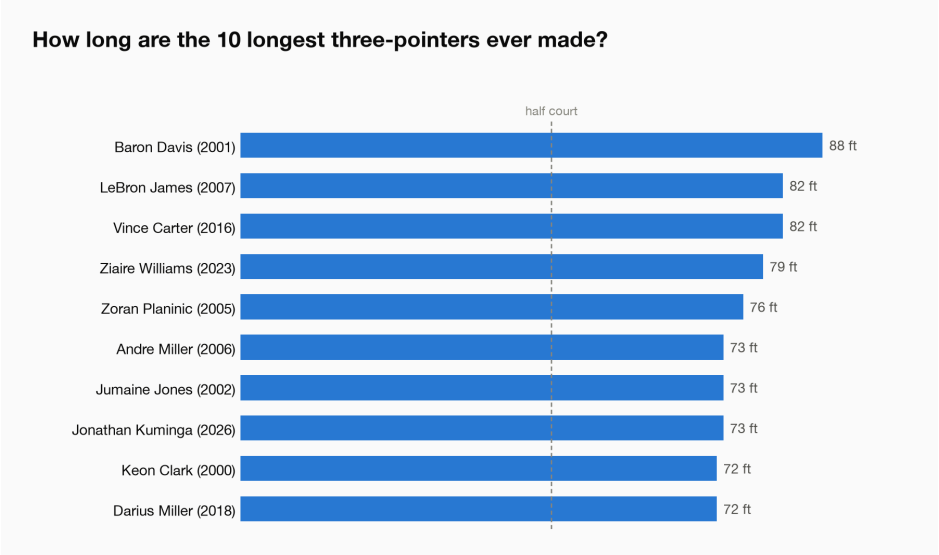
Part V - Data Cleaning

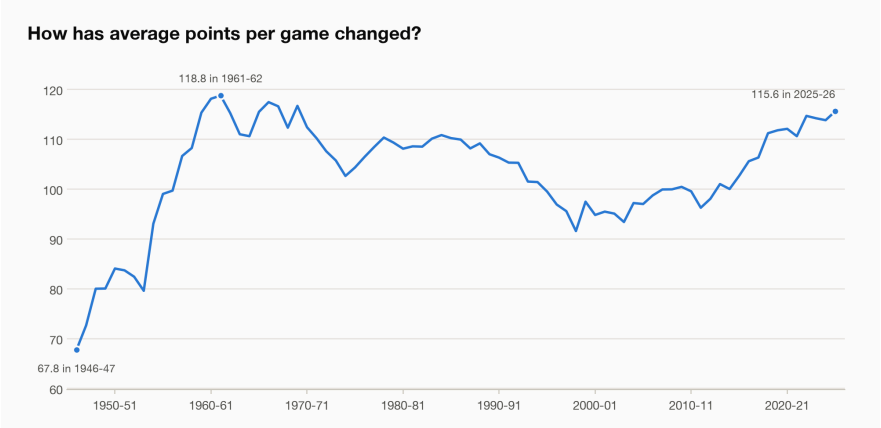
We expect only to do minimal data cleaning in this project because we are fetching the data directly from pre-cleaned sources like the nba_api Python library or curated Kaggle datasets. Instead of raw scraping from NBA.com or other sources. We expect to use python as the tool to process the data. The processing pipeline will mainly fetch the data and cache the data locally in the form of Parquet or CSV. We also may need to process a few data to handle some edge cases. For example, normalizing the shot chart x-y coordinates. Other than that, we believe the rest data should be directly ready for visualization

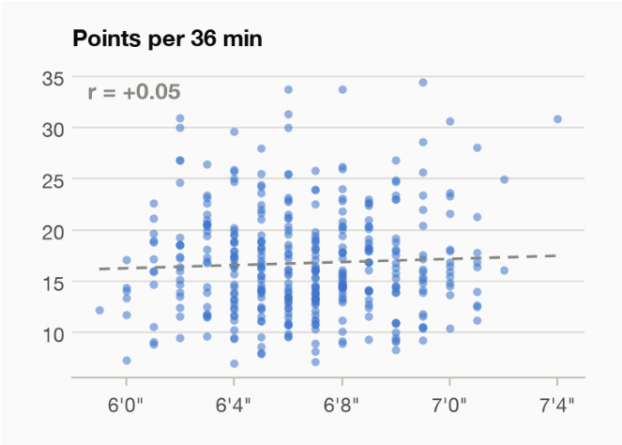
Data Exploration

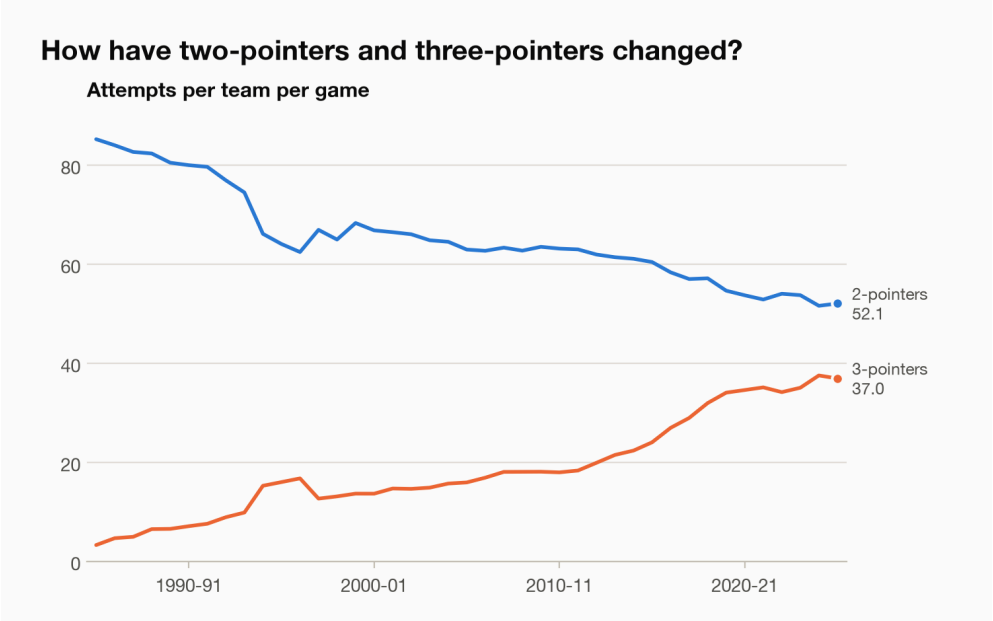
Question 1 not doable, can not get location data for the fouls
Question 3 not doable, no exact time range currently to perform the analysis to player's performance

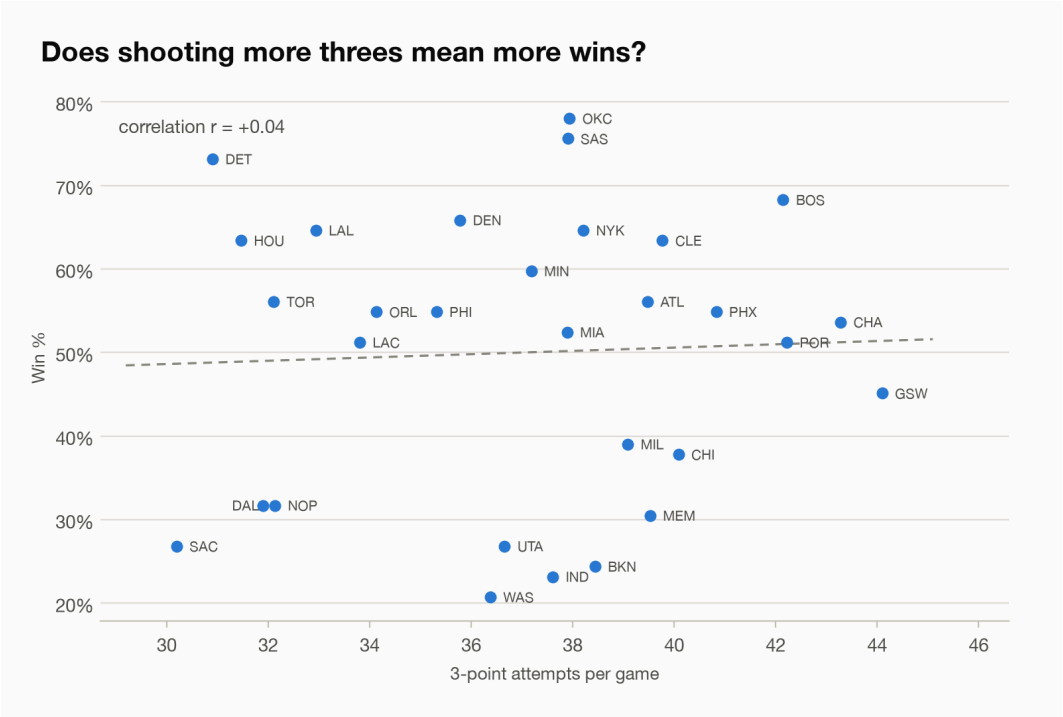
Part I - Preliminary Visualization

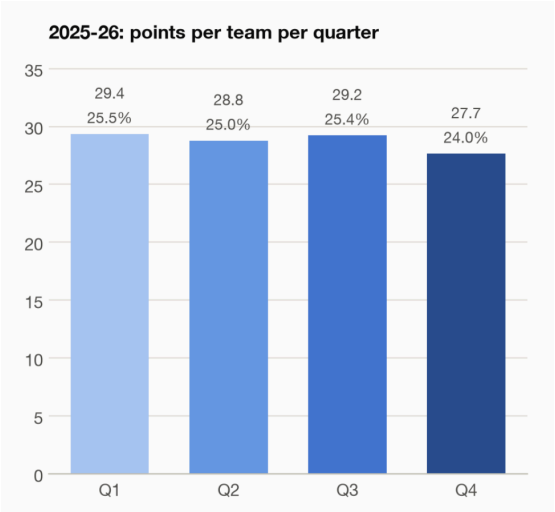


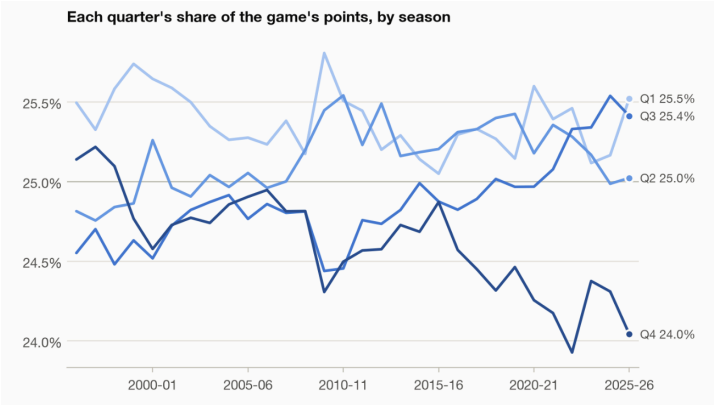



Points per 36 min
r = +0.05
35
30
25
20
15
10
6'0"
6'4"
6'8"
7'0"
7'4"

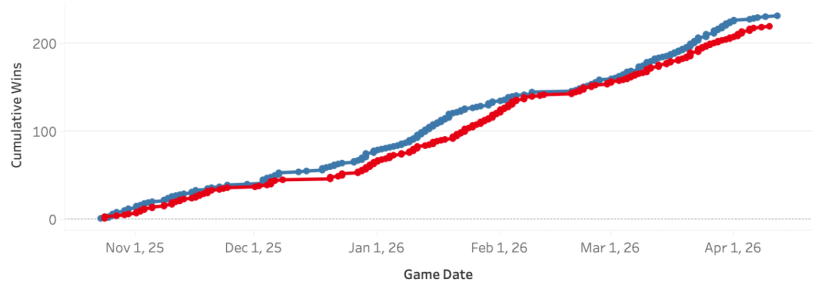

How have two-pointers and three-pointers changed?
Attempts per team per game
80
60
40
20
0
1990-91
2000-01
2010-11
2020-21
2-pointers
52.1
3-pointers
37.0


Does shooting more threes mean more wins?
correlation r = +0.04
Win %
80%
70%
60%
50%
40%
30%
20%
30
32
34
36
38
40
42
44
46
3-point attempts per game
OKC
SAS
DET
BOS
DEN
HOU
LAL
NYK
CLE
MIN
TOR
ORL
PHI
ATL
PHX
CHA
LAC
MIA
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MEM
SAC
UTA
BKN
IND
WAS

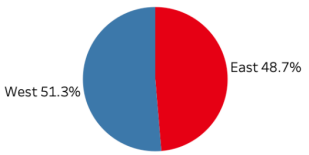


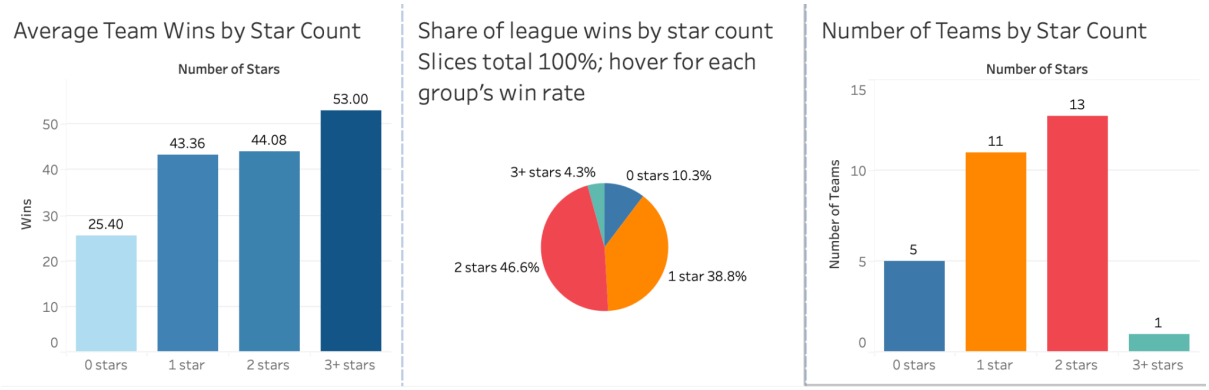


East versus West | cumulative interconference wins, 2025–26
Each dot is one win. East = red · West = blue

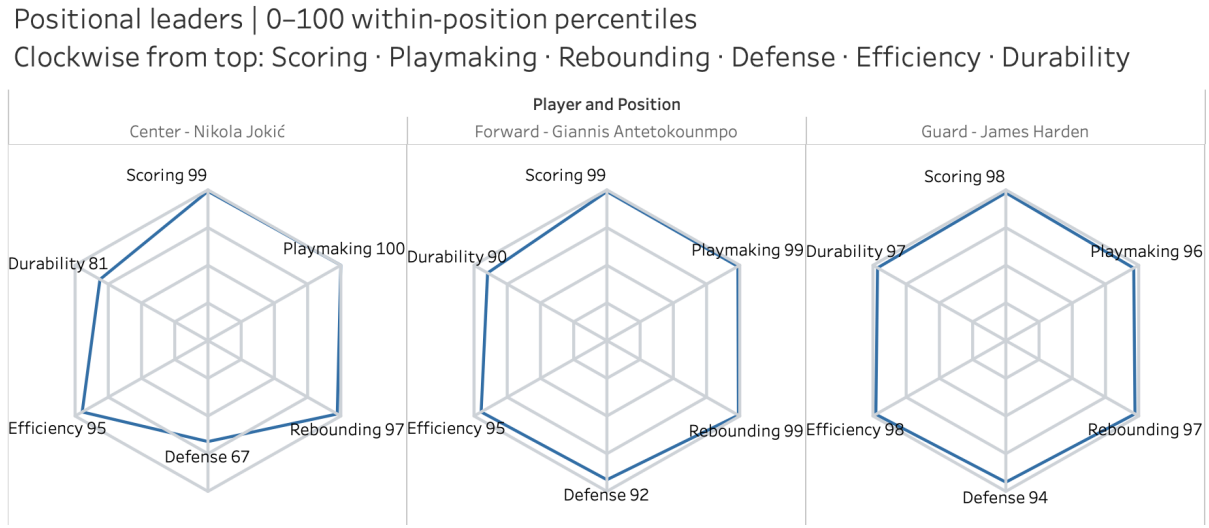


Interconference Win Share (452 East–West Games)

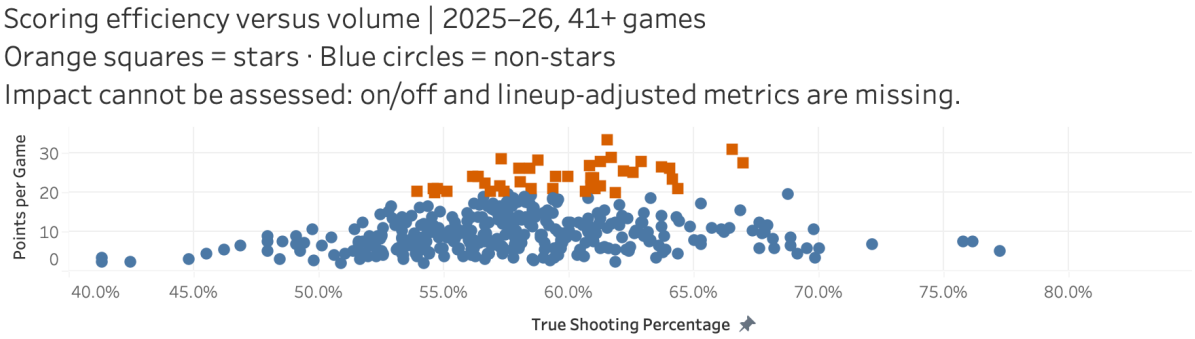




Questions:



Players need at least 300 games in the 1996–97 to 2025–26 coverage window for this metrics. We measure by the percentile.
How would profiles change using peak seasons instead of accumulated career statistics?



Part II - Interesting Findings:

1. Note that the longest three-pointer is almost from the end of the court -Qixiang Fan
2. The top 10 longest three-pointer all happened after 2000 -Qixiang Fan
3. The points per game keep rising for about 20 years and start to decline to about 90 at around 1996 and then back to rising. -Qixiang Fan
4. Player's height seems to have little impact to the points -Qixiang Fan

5. Note that the players in NBA are already been consider very tall (start from about 5’9’’) - Kunhao Ni
6. However, it been shown that there is a clear darker area in between 6’4’’ and 6’8’’
7. Number of 3-points shots doesn’t affect the team’s performance - Kunhao Ni
8. Q4 is always to have the least points scored across 4 quarters - Kunhao Ni
9. For 2025-26 season, the match-wins between the East and West games are almost the same. - Zhaorui Ding
10. Defining a “star” to be 20 points on average for at least 41 games, teams with 0 star in the team has significantly lower game wins than team with at least 1 star. - Zhaorui Ding
11. Most teams has 1-2 stars. (24 out of 30) - Zhaorui Ding
12. When finding the best players, Harden’s six components are consistently high within the guard comparison group. - Yurun Zhu
13. Center’s best player seems to have much lower average abilities. - Yurun Zhu
14. Across the 361 included players, average player TS% is approximately 59.8% for stars versus 57.8% for non-stars, but stars have a smaller variance for true shooting. - Yurun Zhu

Part II - Additional Questions:

1. Players need at least 300 games in the 1996–97 to 2025–26 coverage window for this metrics. We measure by the percentile. How would profiles change using peak seasons instead of accumulated career statistics?
2. Further questions: does east and west have similar basketball skills in the past? Or has it grown to be similar?
3. Does scoring efficiency translate into better team outcomes?
4. We are using scoring to define the stars. How would it change if they were defined using awards, efficiency, or other performance?
5. Where exactly on the court are these top 10 longest three-pointers?
6. How can we measure the player’s “performance”?
7. What caused the decline from 1960s to 1990s, is it because of the playstyle or the changes of rule?

Week 4 - Storyboard

Part I - Summary

The insight we picked for the storyboard is:

> **Although the NBA has clearly shifted toward perimeter-oriented offense, shooting more three-pointers alone does not lead to more wins—how efficiently and strategically teams use the three-point shot matters more than attempt volume.**

A popular argument among NBA fans is that the league has shifted from paint-centered offense toward perimeter-oriented play. However, successful teams do not simply take more

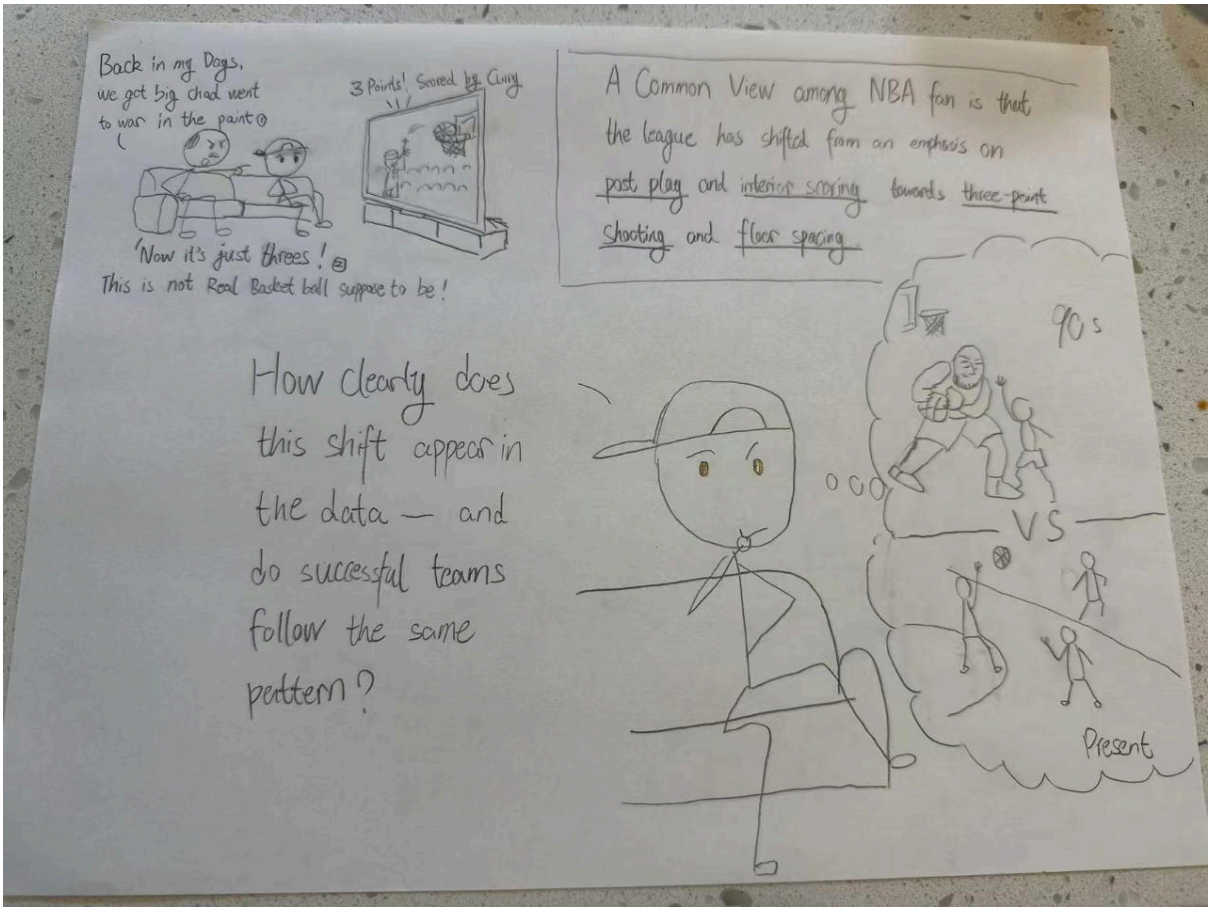
three-point shots. They appear to combine efficient long-range shooting with multiple scoring threats and a balanced offensive system.

We chose this insight because many fans describe modern basketball as being all about three-pointers. Our data supports part of this argument, as three-point attempts have increased greatly over time. However, our analysis does not show a strong relationship between three-point attempt volume and team winning percentage.

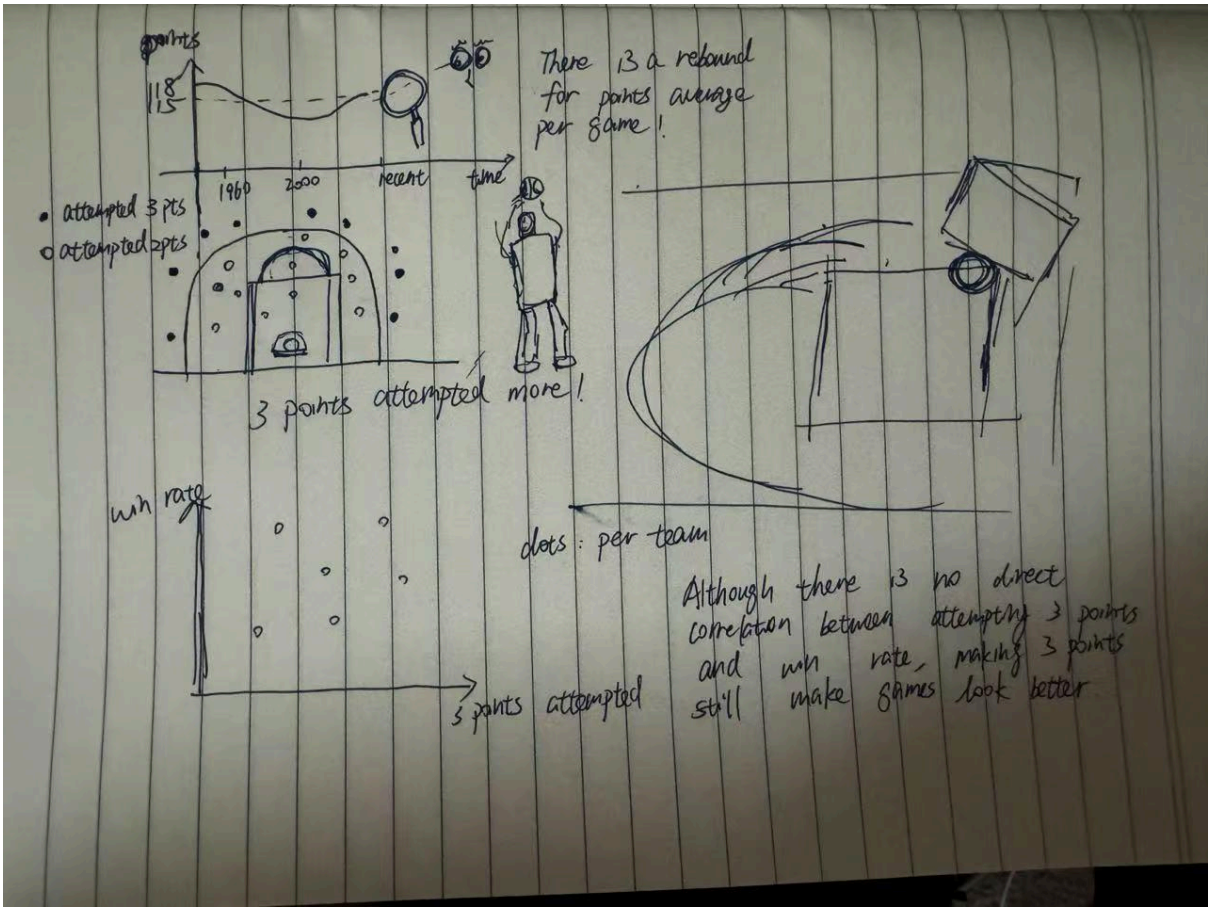
Therefore, our storyboard explores two related questions: How does the data show the NBA's shift toward perimeter-oriented basketball, and what other factors may help explain why some teams succeed in this new era?

Part II - Storyboard

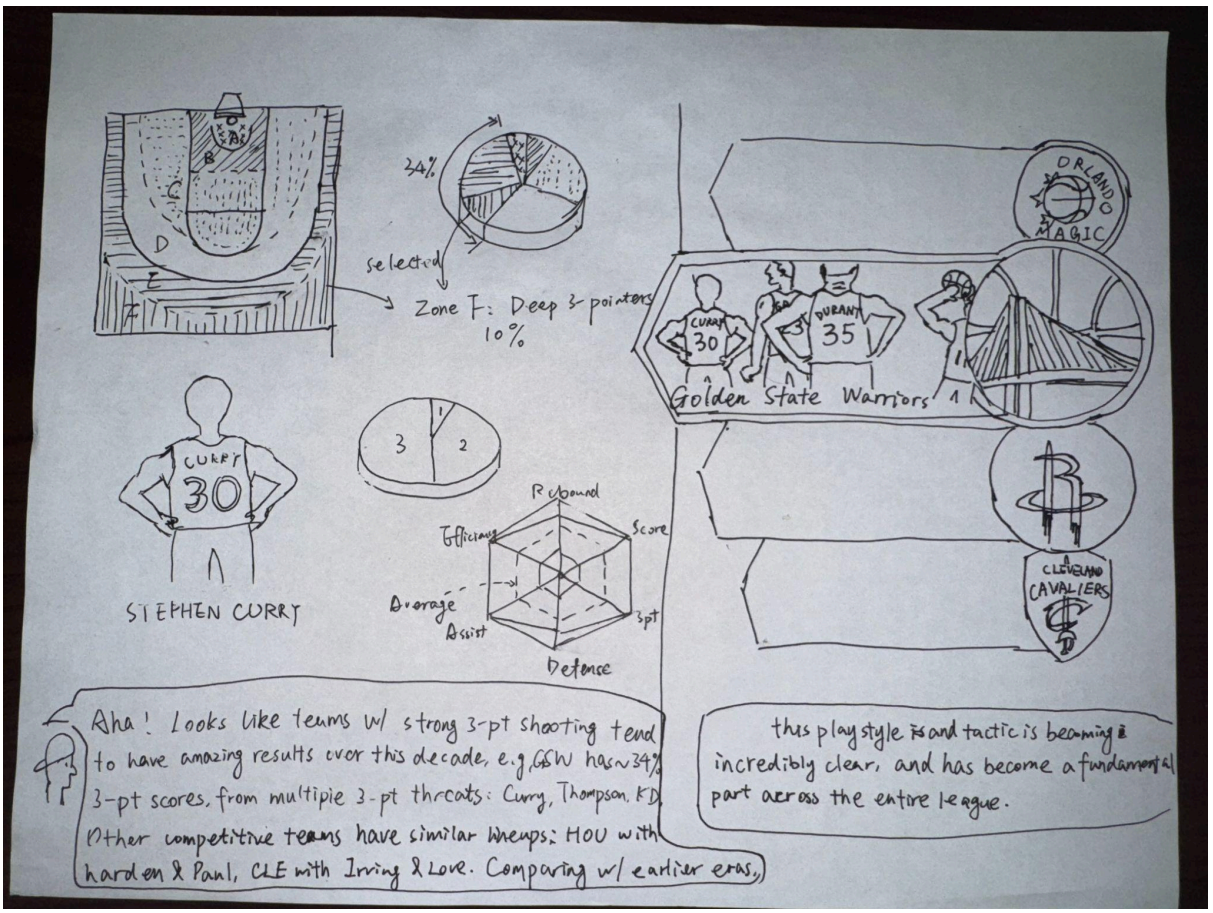
Hook:



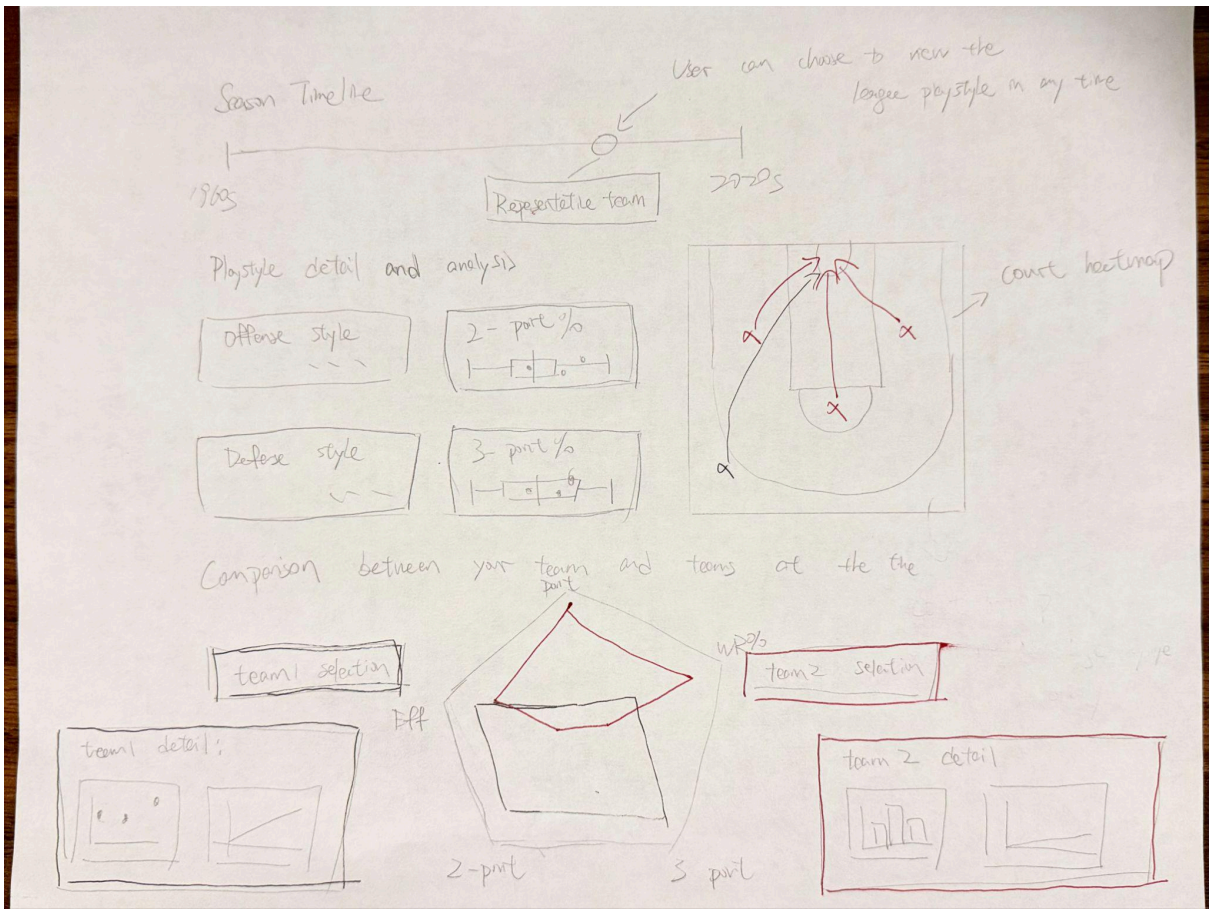
Rising Inslights:



Main Message:


34%
selected
Zone F: Deep 3-pointers
10%
CURRY
30
STEPHEN CURRY
3
2
Rebound
Score
Efficiency
Average
Assist
3pt
Defense
ORLANDO MAGIC
CURRY 30
DURANT 35
Golden State Warriors
CLEVELAND CAVALIERS
Aha! Looks like teams w/ strong 3-pt shooting tend to have amazing results over this decade, e.g. GSW has ~34% 3-pt scores, from multiple 3-pt threats: Curry, Thompson, KD. Other competitive teams have similar lineups: HOU with Harden & Paul, CLE with Irving & Love. Comparing w/ earlier eras,
this playstyle and tactic is becoming incredibly clear, and has become a fundamental part across the entire league.

Solution:

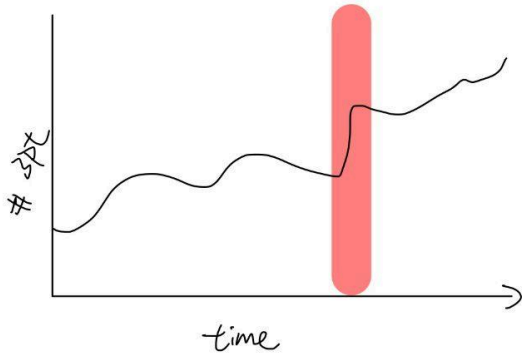


Feedback:

- More Evidence
- Need to justify, and test statement
- Do not use “what-cause -what”
- Make sure to take into account all variables: era, race...
- Need specific question
- winrate vs attempt rate vs efficiency

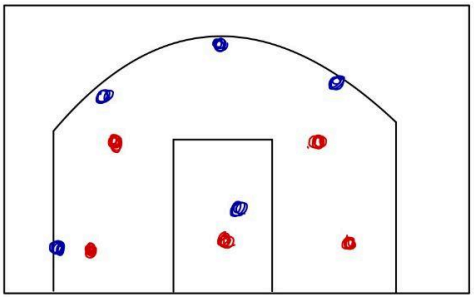
Week 5 - Sketches

Qixiang Fan:



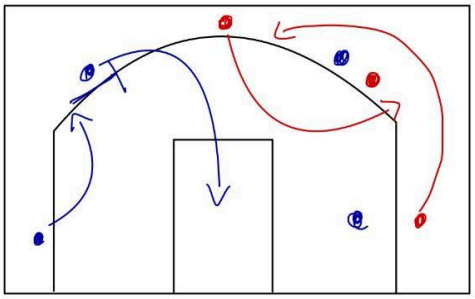
Signifiant 3pt rule change.

What happened to let the 3pt era start



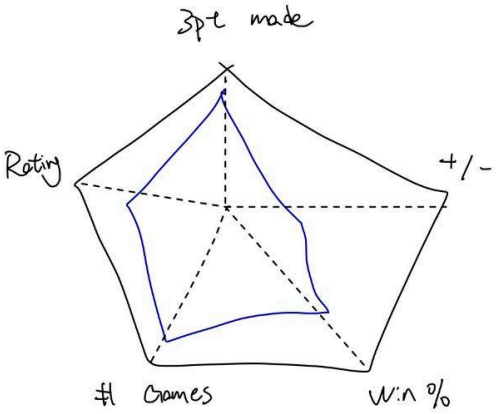
[1] 2020s
[2] 2000s

How the team chage their denfensiy stragey in 3pt Era



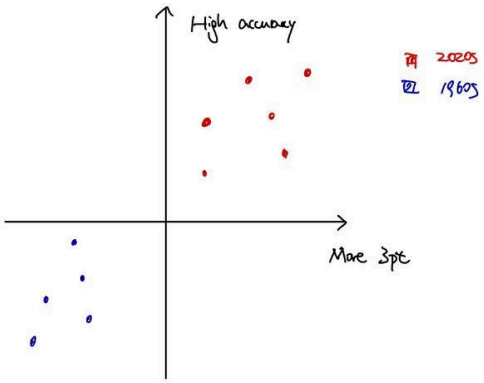
[1] 2020s
[2] 2000s

How the team chage their atteckiy stragely in 3pt Era

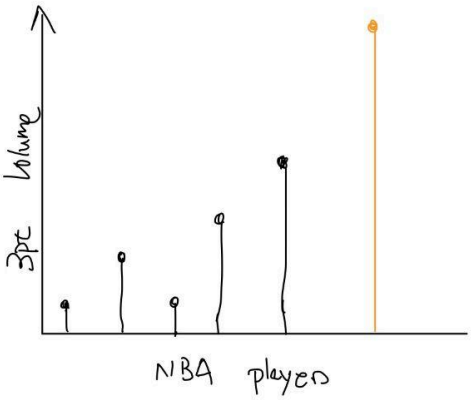


[1] player>

What we think are the key data to reflect the performance, to show the best player in this scale

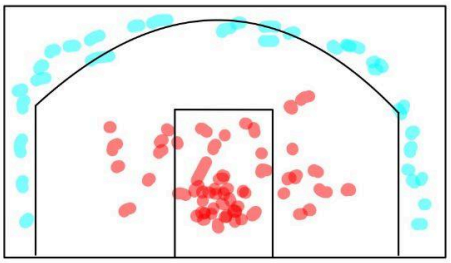


Visualize the shift from 1980 till now



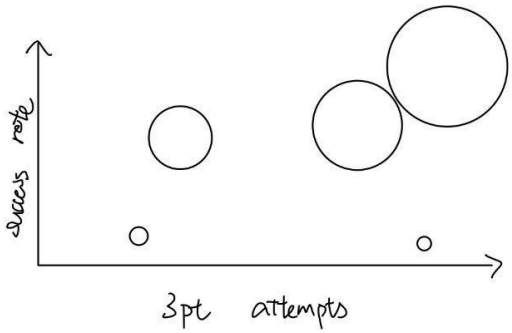
Stephn Curry

Visualize the performone of curry since he is the key player in 3point era



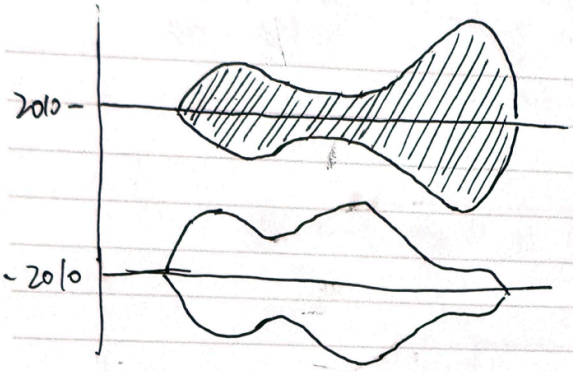
2020s shots
1980s shots

The shooting range is become much deeper

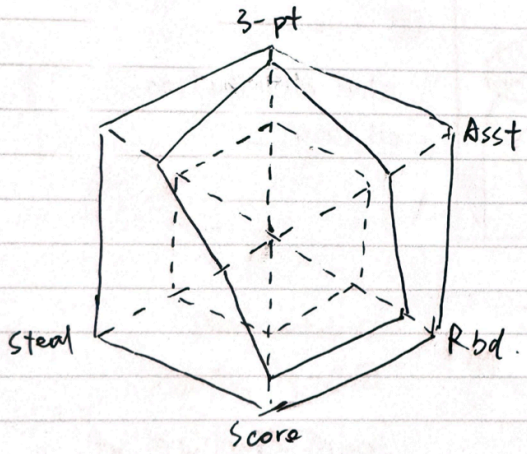


Each bubble is a team, size order by season ranking, show the 3pt performone highly affect the ranking.

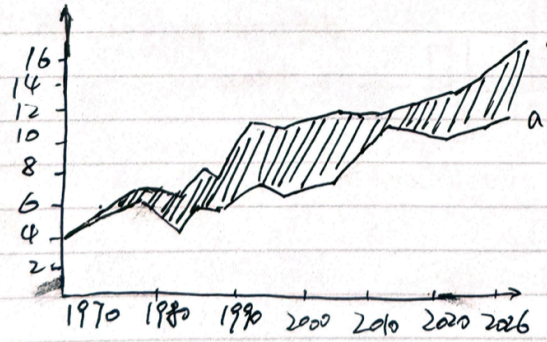
Zhaorui Ding

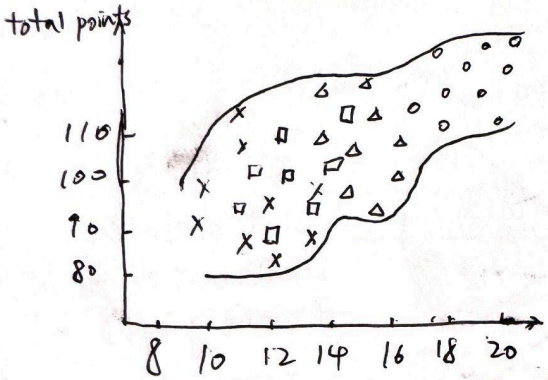


shot distance distribution (violin graph)



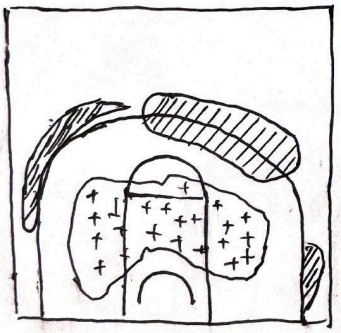
radar chart for single player/ comparing 2+ players





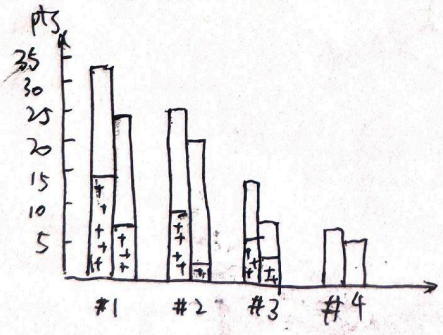
○ 2016-2026
△ 2006-2016
□ 1996~2006
× 1986~1996

champion teams' points / 3-pt correlation



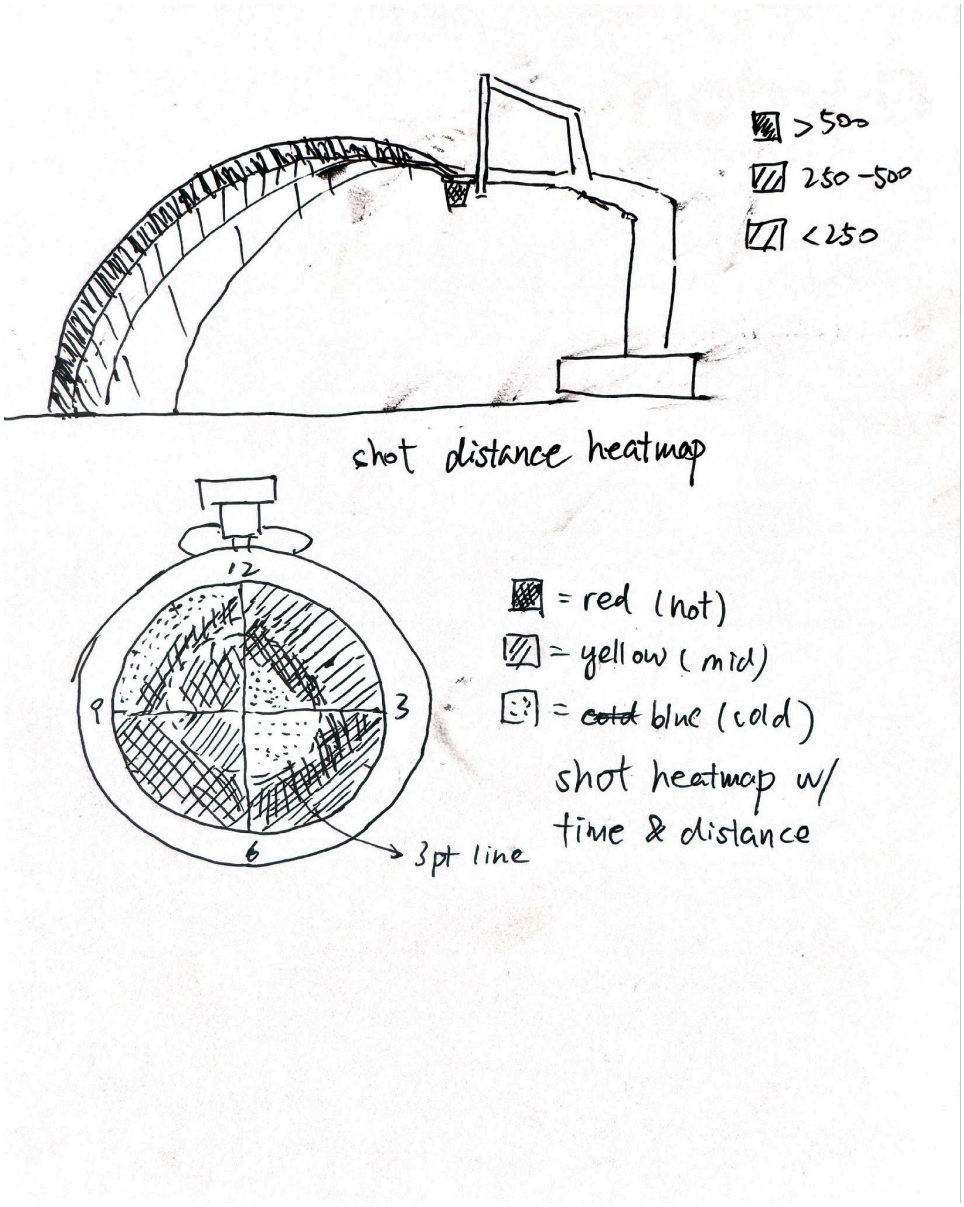
▨ 2010-2026
⊞ -2010

shot distribution on court



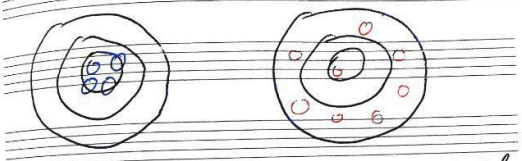
□ total points
⊞ 3-pt points.

point structure of different players in a team.


>500
250-500
<250
shot distance heatmap
12
9
3
6
= red (hot)
= yellow (mid)
= ~~cold~~ blue (cold)
shot heatmap w/ time & distance
3pt line

Jun Woo Beck:

#1. Databoard: volume VS accuracy (Hook)

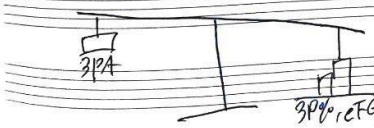


Team A: Fewer, tight Team B: more, scattered

Msg: shooting more 3 ≠ shooting well
→ which one win?

Enc: dot = 3PA, distance from center = miss
color = team

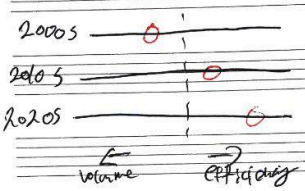
#2 win scale (Rising insight)



Msg: Efficiency → associate with win rate than 3PA volume

Enc: weight = win rate

#3 Volume vs Efficiency by era (Rising insight)



Msg: winning shifts from volume to efficiency over t

Enc: dot = metric related to win rate, dotted line =
left = 3PA, right = 3P%, rainbow

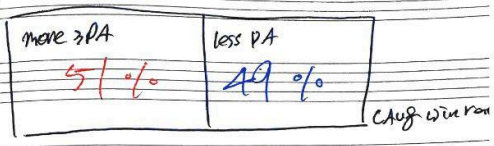
#4 3PA win link over time (Rising insight)



Msg: Hypothesis: 3PA benefit for [illegible] once everyone does it.

Enc: height = r(3PA, win %), x = season, d

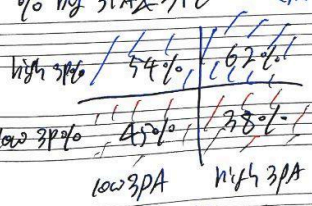
#5 win rate compare (by 3PA number) (Rising insight)



Msg: number of 3PA has no effect on win rate

Enc: Number = Average win rate

#6 win % by 3PA & 3P% (Main msg)



Msg: volume pays off only with accuracy

Enc: team group, num/color
= Avg win %.

#7. champion's number (Main msg)

2014	2017	2021	2024
3PA 21.4	3PA 31.2	3PA 37	3PA 42
3P% 36.0	3P% 34.2	3P% 33.1	3P% 32.8

Msg: champion's 3PA went up, but win rate stays

Enc: Card = champion season,

#8. shooters / team (main msg)

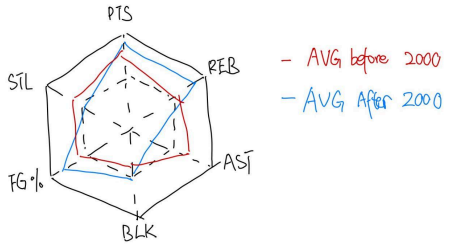


Msg: winning team has more shooters

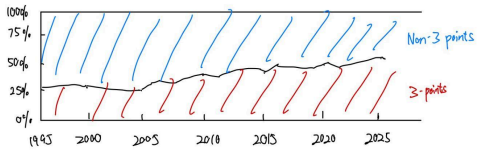
Enc: jersey = starter, 🏀 = shooter

Kunhao Ni

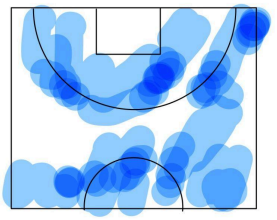
① Hexagonal Pinning



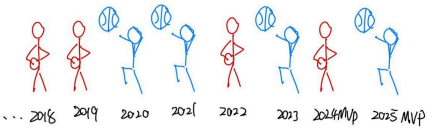
② Stacked Area Chart



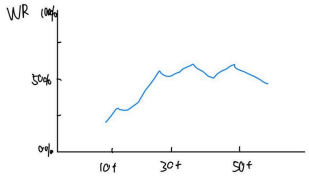
③ Scoring Position Heat Map



④ MVP Player Positions By Seasons



⑤ Winning Rate vs 3-Points Shots

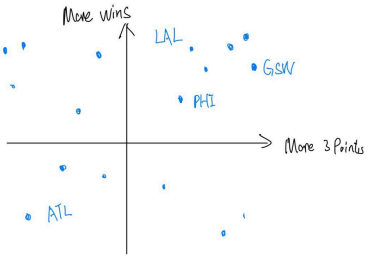


⑥ Scoring Composition Pie Chart

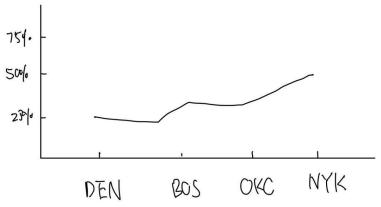


Before vs Now

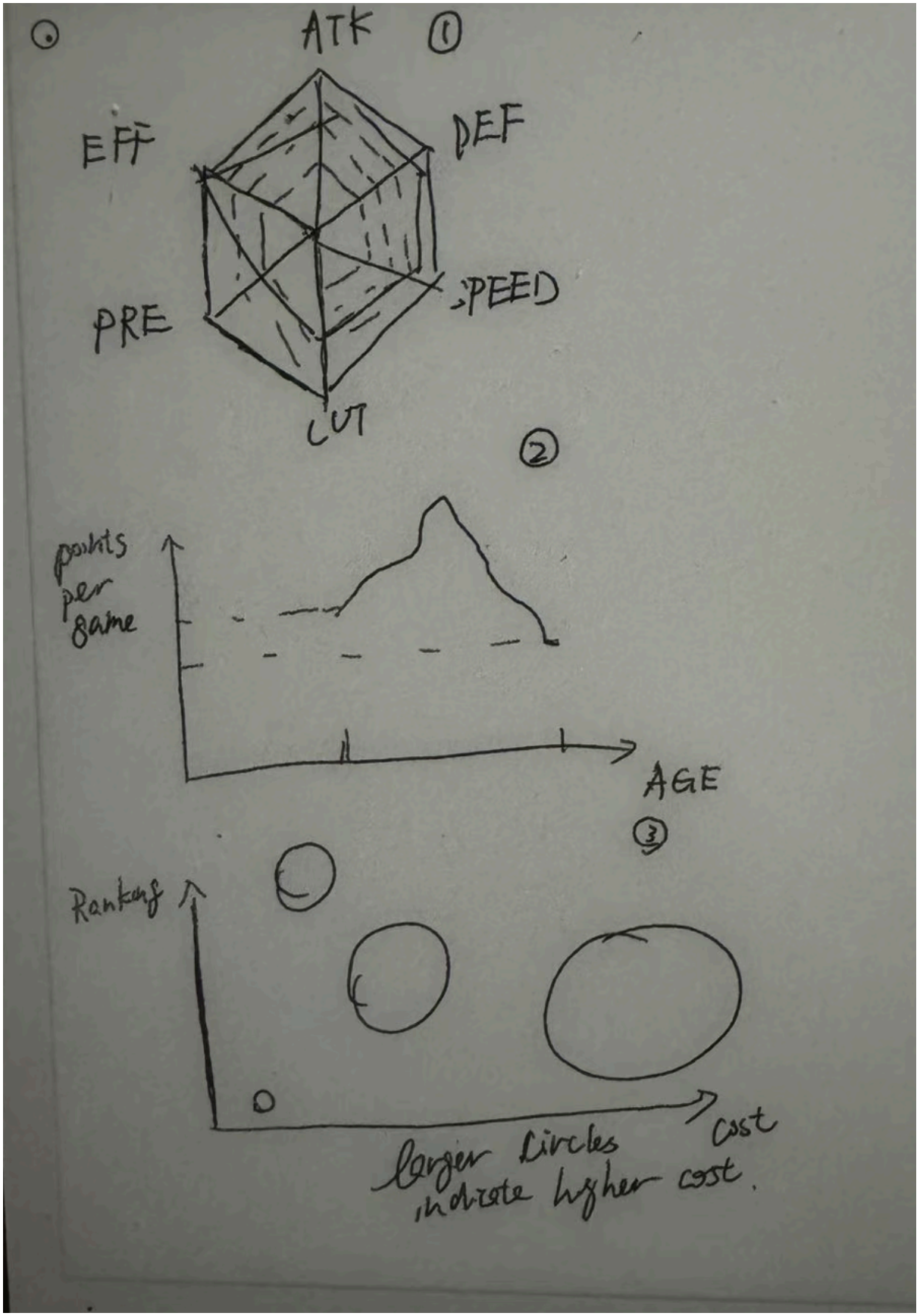
⑦ Team's Win and 3-Points

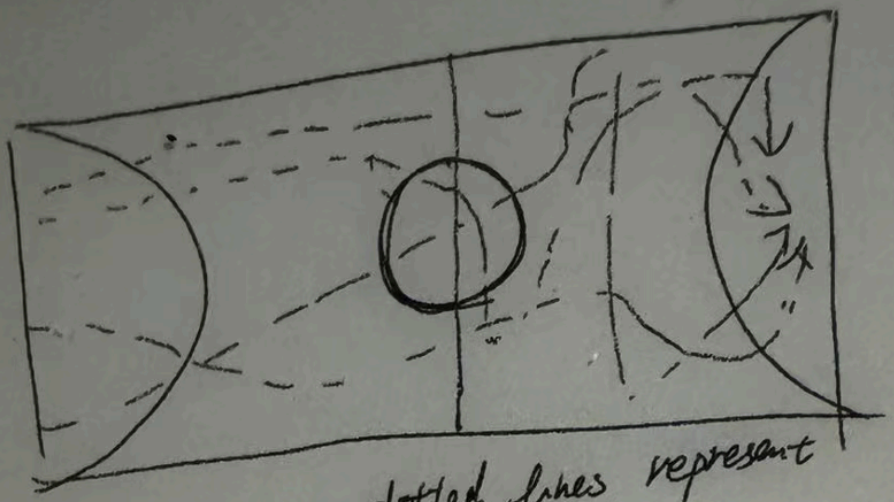


⑧ Champions' 3-points Rate over years



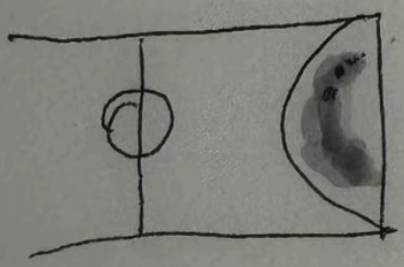
Yurun Zhu:


ATK
①
EFF
DEF
SPEED
PRE
CUT
②
points per game
AGE
③
Rankings
cost
larger circles indicate higher cost.



④

dotted lines represent attack routes



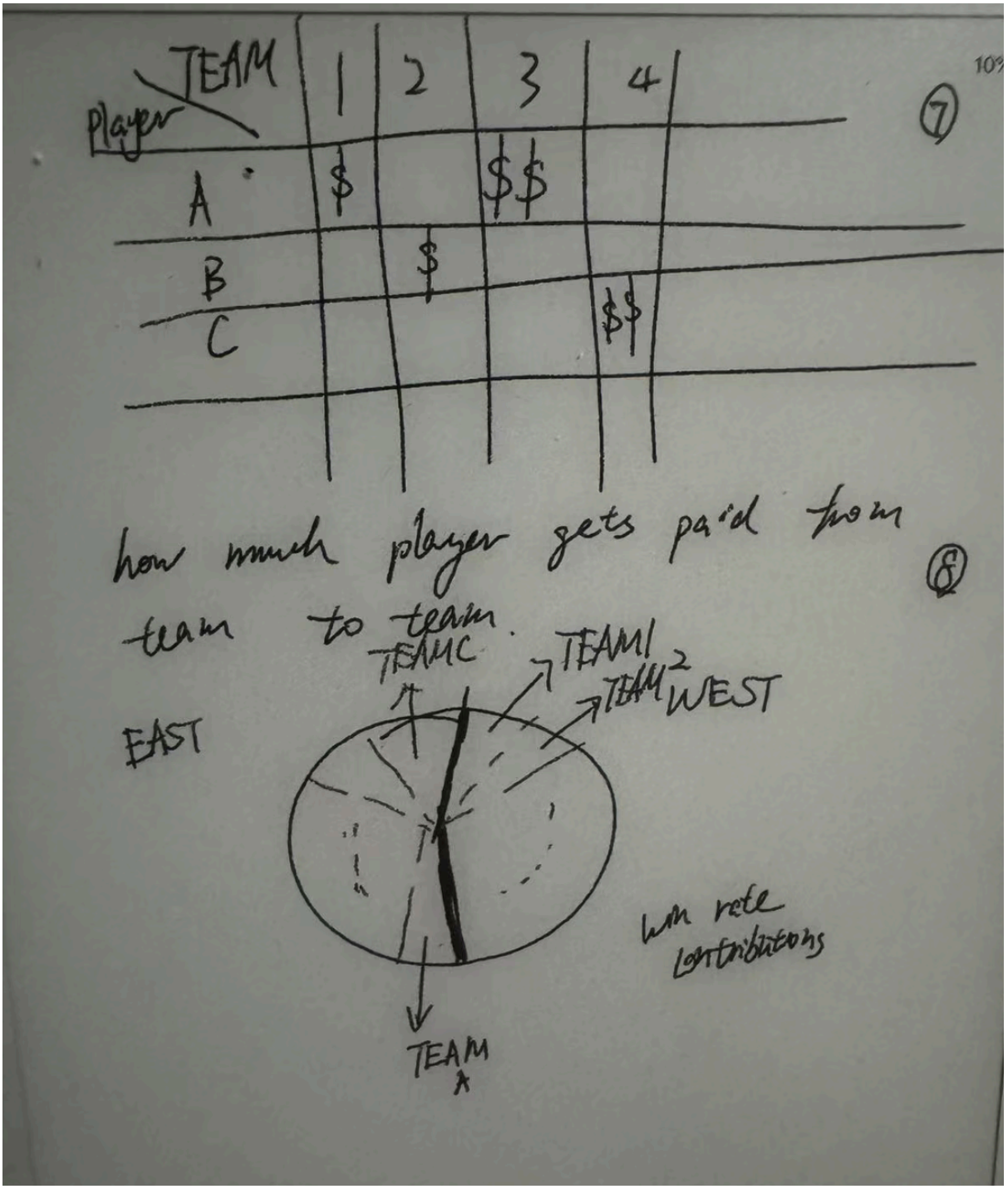
layup position ⑤

⑥



Pause time for 2 teams

Time



Week 6 - Decide

Author initials: QF = Qixiang Fan; YZ = Yurun Zhu; KN = Kunhao Ni; ZD = Zhaorui Ding; JB = Jun Woo Beck.

Message Definitions:

Message ID	Message / Question
M1	How has three-point shooting changed over time?
M2	How has the spatial distribution of shots changed?
M3	How have offensive and defensive strategies changed across eras?
M4	How are three-point volume and accuracy related to team success?
M5	How do players differ in performance and scoring profiles?
M6	How is scoring distributed among players within a team?
M7	How is the number of high-scoring stars related to team success?
M8	What events or rule changes triggered the rise of the three-point era?

Qixiang Fan:

Sketch ID	**Message / Question**
1	M8 - What rule change triggered the sharp rise in three-point shooting?
2	M3 - How did teams change their defensive positioning between the 2000s and the 2020s?
3	M3 - How did teams change their offensive movement and shot selection between the 2000s and the 2020s?

4	M5 - Which player performs best across 3PT made, +/-, win %, games played, and rating?
5	M1 - How did three-point attempts and accuracy shift from the 1960s to the 2020s?
6	M5 - How does Stephen Curry's three-point volume compare to other NBA players?
7	M2 - How does shot distance distribution differ between the 1960s and the 2020s?
8	M4 - How are a team's three-point attempts and success rate related to its season ranking?

Votings: ZD7,ZD8,KN1,KN4,JB8

Zhaorui Ding

Sketch ID	**Message / Question**
1	M2 - How does shot distance distribution differ between 2 different years
2	M5 - Comparison of different players in 6 dimensions
3	Other - How does average and champion teams' 3-pt shot (and their difference) changed over time
4	Other - Champion teams' total points vs 3-pt correlation and classification by eras.
5	M2 - How has the spatial distribution of shots changed?
6	M6 - How is scoring (and 3-pts) distributed among players within a team?
7	M2 - Shot distance distribution heatmap of a team or a season
8	M3? - Shot distribution heatmap of a team in a game, spatial and temporal.

Votings: QF3, JB6, KN1, KN2, YZ6

Jun Woo Beck:

#	Message / Question
1	M4 - Does shooting more threes mean shooting them better? (volume vs accuracy)

2	M4 - Which is more correlated with win %: 3PA or 3P%?
3	M4 - How has the relative importance of 3PA vs 3P% for winning changed across eras?
4	M4 - How has the correlation between 3PA and win % changed over time?
5	M4 - Do teams shooting more threes than the league average win more, in each era?
6	M4 - How does win % differ by high/low 3PA and high/low 3P%?
7	Other - How have champion teams' 3PA and 3P% changed over time?
8	M6 - Do winning teams have more reliable 3-pt shooters?

Votings: QF1,QF8, ZD2, KN4, YZ1

Kunhao Ni:

Sketch ID	**Message / Question**
1	M3 – How do average scoring, rebounding, assists, blocks, steals, and shooting accuracy differ before and after 2000?
2	M1 – How has the share of scoring from three-pointers versus other sources changed over time?
3	M2 – Which court regions contribute the largest share of scoring?
4	M5 – How have the playing positions of MVP winners changed across seasons?
5	M4 – How does team winning percentage vary with three-point shot volume?
6	M1, M2 – How has scoring composition across shot types changed

	between earlier and recent eras?
7	M4 – How do teams compare in three-point shooting volume and season wins?
8	M1, M4 – How has championship teams' reliance on three-point shooting changed over time?

Votings: QF1 QF5 ZD2 YZ4 ZD8

Yurun Zhu:

Sketch ID	**Message / Question**
1	M5 - Compaison of different players in 6 dimensions
2	M5 - Comparisons of players by age
3	M7 - How good is a team over how much they cost
4	M2 - Attack route
5	M2 - Spatial distribution of layup
6	Other - pause time in the game
7	M7 - Player served in team and their cost
8	Other - east/west union combat and team contributions to win rate

Votings: QF1,QF8, ZD2, JB2, KN7

Affinity Diagram:

Sketch ID(s)	**Message ID(s)**	**Author(s)**	**Merged Question / Design**	**Total Votes**
KN2, KN6	M1, M2	KN	How has the composition of scoring changed across eras? — Proportion charts comparing scoring sources	1

ZD3, JB7, KN8	M1, M4	ZD, JB, KN	How has championship teams' three-point shooting changed over time? — Champion-team trends and comparisons	0
QF5, JB1	M1, M4	QF, JB	How are three-point volume and accuracy related? — Volume–accuracy comparisons, with era grouping where available	1
QF7, ZD5, KN3	M2	QF, ZD, KN	Where on the court are shots or scoring concentrated, and how do these patterns differ across eras? — Court-based distribution maps	0
ZD1, ZD7	M2	ZD	How does shot distance distribution vary across teams or seasons? — Distance distributions and heatmaps	1
ZD8	M2	ZD	How does shot distribution vary by court location and game time? — Spatial–temporal heatmap	2
YZ5	M2	YZ	Where are layups concentrated? — Layup-location map	0
QF2	M3	QF	How has defensive positioning changed across eras? — Defensive-position maps	0
QF3, YZ4	M3	QF, YZ	What offensive routes do teams use, and how do these differ across eras? — Court diagrams with movement paths	2
KN1	M3	KN	How do average team statistics differ before and after 2000? — Era-comparison radar chart	2

QF8, KN7	M4	QF, KN	How does team three-point shooting relate to season success? — Team scatterplots, with accuracy or ranking included where available	3
JB2, JB3	M4	JB	Is three-point volume or accuracy more closely associated with winning, and does this differ across eras? — Comparisons of associations with winning	1
JB4	M4	JB	How has the correlation between three-point volume and winning changed over time? — Correlation timeline	0
JB5, KN5	M4	JB, KN	How does winning percentage differ with three-point shot volume? — Winning-percentage comparisons by volume	0
JB6	M4	JB	How does winning percentage differ across high/low three-point volume and accuracy groups? — Four-group comparison	1
ZD4	M4	ZD	How is championship teams' total scoring related to three-point shooting across eras? — Era-grouped scatterplot	0
QF4, ZD2, YZ1	M5	QF, ZD, YZ	How do players compare across multiple performance dimensions? — Player-comparison radar charts	4
QF6	M5	QF	How does Stephen Curry's three-point volume compare with other players? — Player-volume comparison	0

KN4	M5	KN	How have MVP winners' playing positions changed across seasons? — Position timeline	2
YZ2	M5	YZ	How does player scoring performance vary with age? — Age–scoring trend	0
ZD6	M6	ZD	How are total and three-point scoring contributions distributed within a team? — Player scoring bars	0
JB8	M4, M6	JB	Do teams with more reliable three-point shooters win more? — Shooter-count comparison by team success	1
QF1	M1, M8	QF	What events or rule changes coincided with changes in three-point shooting? — Annotated timeline	3
YZ3	M9	YZ	How is team spending related to team ranking? — Cost–ranking bubble chart	0
YZ7	M9	YZ	How do player salaries vary across teams? — Player–team salary matrix	1
YZ6	M10	YZ	How do two teams use timeouts during a game? — Timeout timelines	1
YZ8	M11	YZ	How do Eastern and Western Conference teams contribute to winning performance? — Conference and team composition chart	0
Total			**40 sketches organized into 27 groups**	**25**

Voting:

Votes	Sketch ID(s)
3	QF1, ZD2
2	QF8, KN1, KN4, ZD8
1	QF3, QF5, ZD7, JB2, JB6, JB8, KN2, KN7, YZ1, YZ4, YZ6, YZ7
0	All remaining sketches

After grouping all of our sketches by message and voting as a team, we selected the 6 designs that have most votes and best fit to our story. During the voting, we noticed that some sketches we created can not supported by our data cannot, such as player movement routes, and there are also some skectch ideas outside our main question, such as salaries, timeouts, and conference comparisons. We only chosed the designs that have a strong support to the story and our audience of NBA fans can read easily, including line charts, scatterplots, radar charts, and court diagrams. We first vote for the line plot which is a timeline of three-point attempts annotated with key rule changes, to show when the shift happened and what coincided with it. Then the heatmap in order to show the audience where on the court shots have moved over time, and compare how overall team statistics differ between earlier and recent eras. The main focus is the scatter plot which directly relates three-point volume and accuracy to season success, it directly answer the story we had. Finally, we favored the radar chart and timeline of MVP positions to move the scope from teams to individual players with a multi-dimensional player comparison and a look at how the positions of MVP winners have changed.